

An Update to “Polynomial Hashing over Prime Order Fields”

Kaushik Nath¹ and Palash Sarkar¹

¹Indian Statistical Institute, 203, B.T. Road, Kolkata, India 700108
Emails: kaushik.nath@yahoo.in, palash@isical.ac.in

July 1, 2025

Abstract

New state-of-the-art assembly implementations show that **BRWHash** is consistently faster than **polyHash** and both *t*-**BRWHash** and *d*-**2LHash** for all message lengths and for both the primes $2^{127} - 1$ and $2^{130} - 5$.

Keywords: almost XOR universal hash function, polynomial hash, BRW hash.

1 Introduction

Various polynomial hashing strategies over prime order fields were systematically studied in [1]. The hash function **polyHash** is the usual polynomial based hashing. The hash function **BRWHash** was proposed in [1] and is based on Bernstein-Rabin-Winograd (BRW) polynomials. Two other hash functions, namely *t*-**BRWHash** and *d*-**2LHash**, were proposed in [1]. These two hash functions combine BRW polynomials with the usual polynomials.

The primes considered in [1] were $2^{127} - 1$ and $2^{130} - 5$. For both the primes, [1] reported implementations of all of the above four hash functions in assembly language targeted towards the **x86_64** architecture. Based on the experimental results arising from the implementations, it was observed that for short messages **polyHash** is the fastest among the four hash functions, and starting from messages longer than a few blocks, *d*-**2LHash** is the fastest among the four hash functions. Accordingly, a hash function *d*-**Hash** was proposed which applied **polyHash** for short messages and switched to *d*-**2LHash** once the number of blocks in the message crossed a pre-defined threshold.

The implementations reported in [1] were non-uniform in the sense that the code for the BRW component used for **BRWHash** and *t*-**BRWHash** was different from the code for the BRW component used in *d*-**2LHash**. To get rid of this inconsistency, we have made fresh implementations of **BRWHash** and *t*-**BRWHash** so that the code for the BRW component becomes the same for all the three hash functions.

Timing results obtained from the new implementations show that for both the primes $2^{127} - 1$ and $2^{130} - 5$, **BRWHash** is consistently faster than **polyHash** and is also faster than both *t*-**BRWHash** and *d*-**2LHash** for all message lengths. The new implementations of **BRWHash** and *t*-**BRWHash** as part of this update can be found at the following links.

<https://github.com/kn-cs/BRWHash>

<https://github.com/kn-cs/t-BRWHash>

The implementations of `polyHash` and `d-2LHash` from [1] are available at the following links.

<https://github.com/kn-cs/polyHash>
<https://github.com/kn-cs/d2LHash>

We report timing measurements for the updated code. The timing measurements were taken on a single core of 11th Gen Intel Core i7-1185G7 @ 3.00GHz $\times$ 4 Tiger Lake processor using 31.1 GiB memory. During the experiments, turbo boost and hyperthreading options were turned off. The OS was Ubuntu 20.04.3 LTS and the code was compiled using gcc version 9.4.0. The following flags were used during compilation.

`-march=native -mtune=native -m64 -O3 -funroll-loops -fomit-frame-pointer`

We counted CPU cycles using the microlibrary “`libcpucycles`” (see <https://cpucycles.cr.yp.to/>) through the `amd64-pmc` counter (see <https://cpucycles.cr.yp.to/counters.html>) which requires a 64-bit Intel/AMD platform and Linux `perf_event` interface. The `amd64-pmc` counter accesses a cycle counter through RDPMC and requires `/proc/sys/kernel/perf_event_paranoid` to be at most 2 for user-level RDPMC access. This counter runs at clock frequency of the CPU core.

The hash functions use two kinds of parameters, namely implementation parameters and design parameters. The implementation parameters determine the efficiency of implementation, but do not affect the output of the hash function. On the other hand, the design parameters affect the output of the corresponding hash function. For `polyHash`, the parameter g is an implementation parameter, and for `BRWHash`, the parameter t is an implementation parameter. For `t-BRWHash`, the parameter t is a design parameter, while g is an implementation parameter. Similarly, for `d-2LHash`, the parameter d is a design parameter, while g is an implementation parameter. For more details about the parameters we refer to [1].

In the following tables, we report timing measurements for the different hash functions. The numbers in the tables are cycles-per-byte measurements. Some of the cells have a pair of entries, the first entry of such a pair report the speed when certain powers of the key are pre-computed, while the second entry report the speed when these same powers are computed on-the-fly.

The implementation parameter t of `BRWHash` determines the extent of loop unrolling. The number of multiplications required by `BRWHash` is about half that required by `polyHash`. The timing results, on the other hand, show that the speed improvement of `BRWHash` over `polyHash` with $g = 1$ is more than two-fold. This is due to the loop unrolling determined by the implementation parameter t of `BRWHash`. Note, however, that the speed improvement of `BRWHash` over `polyHash` is lower than a factor of two as soon as g is chosen to be greater than 1. The implementation parameter g of `polyHash` determines the extent of loop unrolling. See [1] for an explanation of the higher-than-expected speed improvement of `polyHash` when g is chosen to be greater than 1.

From the timing results in the tables we note that `BRWHash` is faster than both `t-BRWHash` and `d-2LHash`. Further, the speed improvement of `BRWHash` over `polyHash` is about 20% or more for both the primes. In view of the timing results reported in the tables, we update the recommendation of [1] to suggest the use of `BRWHash` instead of `d-2LHash` for all message lengths and for both the primes $2^{127} - 1$ and $2^{130} - 5$. The speed improvement of `BRWHash1271` over `BRWHash1305` is about 30% or higher for messages longer than a few bytes, and the speed improvement of `BRWHash1271` over `polyHash1305` (which is the well known Poly1305 for byte length messages with appropriate key clamping) is higher. To summarise, among all the hash functions defined in [1], `BRWHash1271` is the fastest, and provides a considerable speed improvement over Poly1305.

# blks	polyHash1271					BRWHash1271			
	$g = 1$	$g = 4$	$g = 8$	$g = 16$	$g = 32$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
1	2.93	2.40, 2.40	2.47, 2.47	2.47, 2.47	2.53, 2.53	2.13, 2.13	2.33, 2.33	2.27, 2.27	2.27, 2.27
2	2.20	1.53, 1.53	1.57, 1.57	1.57, 1.57	1.57, 1.57	1.27, 1.27	1.33, 1.33	1.23, 1.23	1.30, 1.30
3	1.93	1.18, 1.47	1.20, 1.49	1.22, 1.49	1.24, 1.56	0.87, 1.18	0.91, 1.20	0.91, 1.18	0.91, 1.18
4	1.77	1.02, 1.45	1.00, 1.43	1.02, 1.45	1.05, 1.48	1.02, 1.38	0.85, 1.22	0.85, 1.23	0.85, 1.22
5	1.69	0.99, 1.47	0.91, 1.37	0.92, 1.37	0.93, 1.40	0.80, 1.11	0.71, 1.00	0.72, 1.01	0.71, 1.00
6	1.63	0.88, 1.27	0.83, 1.37	0.83, 1.37	0.86, 1.39	0.71, 0.97	0.67, 0.92	0.67, 0.91	0.66, 0.90
7	1.60	0.83, 1.16	0.78, 1.40	0.78, 1.39	0.80, 1.41	0.63, 0.86	0.60, 0.82	0.61, 0.81	0.60, 0.80
8	1.57	0.79, 1.07	0.73, 1.43	0.73, 1.43	0.76, 1.44	0.77, 1.04	0.69, 1.00	0.68, 0.96	0.68, 0.96
9	1.54	0.80, 1.06	0.75, 1.41	0.71, 1.33	0.73, 1.33	0.69, 0.93	0.64, 0.88	0.62, 0.87	0.62, 0.87
10	1.52	0.76, 0.99	0.71, 1.30	0.68, 1.33	0.69, 1.34	0.64, 0.85	0.59, 0.81	0.60, 0.82	0.60, 0.81
11	1.51	0.73, 0.93	0.68, 1.22	0.66, 1.34	0.68, 1.34	0.59, 0.81	0.55, 0.76	0.56, 0.76	0.55, 0.76
12	1.49	0.71, 0.89	0.67, 1.16	0.64, 1.34	0.65, 1.34	0.68, 0.88	0.57, 0.74	0.57, 0.75	0.56, 0.75
13	1.48	0.73, 0.90	0.66, 1.11	0.63, 1.36	0.64, 1.36	0.64, 0.81	0.54, 0.70	0.54, 0.71	0.53, 0.70
14	1.48	0.70, 0.86	0.64, 1.06	0.61, 1.38	0.62, 1.39	0.60, 0.77	0.53, 0.68	0.53, 0.68	0.52, 0.68
15	1.47	0.68, 0.83	0.63, 1.02	0.60, 1.40	0.61, 1.40	0.57, 0.72	0.50, 0.65	0.51, 0.65	0.50, 0.64
16	1.46	0.67, 0.81	0.62, 0.98	0.60, 1.43	0.60, 1.43	0.63, 0.83	0.57, 0.75	0.55, 0.74	0.55, 0.73
17	1.45	0.68, 0.81	0.65, 0.98	0.62, 1.44	0.60, 1.38	0.60, 0.80	0.53, 0.72	0.51, 0.70	0.52, 0.70
18	1.45	0.66, 0.79	0.61, 0.95	0.60, 1.39	0.59, 1.39	0.58, 0.75	0.51, 0.69	0.50, 0.67	0.51, 0.68
19	1.44	0.65, 0.76	0.61, 0.92	0.60, 1.33	0.58, 1.40	0.55, 0.73	0.50, 0.66	0.48, 0.65	0.49, 0.65
20	1.44	0.64, 0.75	0.60, 0.89	0.60, 1.29	0.57, 1.40	0.60, 0.76	0.51, 0.67	0.50, 0.65	0.50, 0.65
21	1.43	0.65, 0.76	0.60, 0.87	0.60, 1.25	0.57, 1.40	0.58, 0.72	0.50, 0.65	0.49, 0.63	0.49, 0.63
22	1.42	0.64, 0.74	0.59, 0.86	0.58, 1.21	0.56, 1.43	0.56, 0.70	0.50, 0.63	0.48, 0.61	0.48, 0.62
23	1.42	0.63, 0.73	0.59, 0.84	0.58, 1.18	0.57, 1.46	0.54, 0.68	0.48, 0.61	0.47, 0.60	0.47, 0.60
24	1.41	0.63, 0.72	0.58, 0.82	0.63, 1.15	0.55, 1.49	0.58, 0.72	0.53, 0.66	0.50, 0.62	0.51, 0.63
25	1.41	0.64, 0.73	0.64, 0.83	0.62, 1.14	0.55, 1.48	0.56, 0.70	0.51, 0.63	0.49, 0.61	0.50, 0.61
26	1.41	0.63, 0.72	0.58, 0.81	0.56, 1.12	0.55, 1.55	0.55, 0.67	0.50, 0.63	0.48, 0.59	0.48, 0.60
27	1.40	0.62, 0.70	0.58, 0.80	0.60, 1.07	0.67, 1.54	0.53, 0.65	0.48, 0.60	0.47, 0.58	0.48, 0.58
28	1.40	0.61, 0.69	0.57, 0.78	0.55, 1.07	0.64, 1.56	0.57, 0.68	0.50, 0.60	0.47, 0.58	0.48, 0.59
29	1.40	0.62, 0.70	0.57, 0.77	0.55, 1.03	0.54, 1.57	0.55, 0.67	0.49, 0.59	0.46, 0.56	0.47, 0.57
30	1.40	0.62, 0.69	0.57, 0.76	0.55, 1.03	0.53, 1.58	0.54, 0.66	0.48, 0.59	0.46, 0.56	0.48, 0.57
31	1.40	0.61, 0.68	0.56, 0.75	0.54, 1.01	0.53, 1.58	0.53, 0.63	0.47, 0.57	0.45, 0.55	0.46, 0.55
32	1.39	0.60, 0.68	0.60, 0.74	0.54, 0.97	0.71, 1.64	0.56, 0.69	0.50, 0.63	0.48, 0.60	0.46, 0.59
33	1.39	0.61, 0.68	0.57, 0.75	0.56, 0.98	0.70, 1.64	0.54, 0.67	0.48, 0.61	0.47, 0.59	0.45, 0.57
34	1.39	0.60, 0.67	0.56, 0.74	0.55, 0.96	0.68, 1.59	0.53, 0.67	0.48, 0.60	0.46, 0.58	0.45, 0.56
35	1.39	0.60, 0.66	0.56, 0.73	0.55, 0.94	0.67, 1.56	0.52, 0.64	0.47, 0.59	0.45, 0.57	0.44, 0.56
36	1.39	0.60, 0.66	0.56, 0.72	0.55, 0.93	0.67, 1.54	0.54, 0.66	0.48, 0.59	0.46, 0.58	0.45, 0.56
37	1.39	0.60, 0.67	0.56, 0.72	0.54, 0.92	0.66, 1.51	0.53, 0.65	0.47, 0.58	0.45, 0.56	0.44, 0.55
38	1.39	0.60, 0.66	0.55, 0.71	0.54, 0.91	0.67, 1.48	0.52, 0.64	0.47, 0.58	0.45, 0.56	0.44, 0.55
39	1.38	0.59, 0.65	0.55, 0.70	0.54, 0.89	0.65, 1.45	0.52, 0.63	0.46, 0.57	0.44, 0.55	0.43, 0.54
40	1.38	0.59, 0.65	0.58, 0.70	0.54, 0.88	0.67, 1.44	0.55, 0.65	0.48, 0.59	0.47, 0.56	0.45, 0.55
41	1.38	0.60, 0.65	0.56, 0.70	0.54, 0.88	0.65, 1.41	0.53, 0.63	0.47, 0.58	0.46, 0.56	0.45, 0.55
42	1.38	0.59, 0.64	0.55, 0.69	0.53, 0.87	0.64, 1.38	0.52, 0.62	0.47, 0.57	0.45, 0.55	0.44, 0.54
43	1.38	0.59, 0.64	0.55, 0.69	0.53, 0.85	0.64, 1.36	0.51, 0.61	0.46, 0.56	0.44, 0.54	0.44, 0.53
44	1.38	0.59, 0.64	0.55, 0.68	0.53, 0.86	0.63, 1.35	0.54, 0.63	0.47, 0.56	0.45, 0.55	0.44, 0.53
45	1.38	0.59, 0.64	0.55, 0.68	0.53, 0.85	0.67, 1.33	0.52, 0.62	0.46, 0.56	0.44, 0.53	0.44, 0.53
46	1.38	0.59, 0.64	0.54, 0.67	0.53, 0.84	0.66, 1.31	0.52, 0.61	0.46, 0.55	0.45, 0.53	0.43, 0.52
47	1.38	0.58, 0.63	0.54, 0.67	0.56, 0.83	0.66, 1.29	0.51, 0.60	0.45, 0.54	0.44, 0.53	0.43, 0.52
48	1.38	0.58, 0.63	0.54, 0.66	0.53, 0.81	0.62, 1.27	0.53, 0.62	0.47, 0.56	0.45, 0.54	0.44, 0.53
49	1.38	0.59, 0.63	0.57, 0.67	0.53, 0.82	0.61, 1.26	0.52, 0.61	0.46, 0.55	0.45, 0.53	0.44, 0.53
50	1.37	0.58, 0.63	0.54, 0.66	0.53, 0.81	0.61, 1.24	0.51, 0.61	0.46, 0.55	0.44, 0.53	0.44, 0.52
100	1.36	0.58, 0.58	0.54, 0.58	0.52, 0.65	0.56, 0.91	0.51, 0.56	0.45, 0.50	0.43, 0.48	0.42, 0.48
150	1.35	0.57, 0.57	0.53, 0.56	0.51, 0.59	0.55, 0.78	0.50, 0.54	0.44, 0.48	0.43, 0.47	0.42, 0.46
200	1.35	0.56, 0.56	0.52, 0.54	0.51, 0.57	0.53, 0.69	0.50, 0.53	0.44, 0.47	0.42, 0.45	0.42, 0.45
250	1.35	0.56, 0.57	0.52, 0.54	0.51, 0.56	0.53, 0.65	0.50, 0.52	0.44, 0.46	0.42, 0.44	0.42, 0.44
300	1.34	0.56, 0.56	0.52, 0.54	0.51, 0.55	0.52, 0.61	0.50, 0.52	0.44, 0.46	0.42, 0.44	0.41, 0.44
350	1.34	0.56, 0.56	0.52, 0.53	0.50, 0.54	0.51, 0.61	0.50, 0.51	0.44, 0.46	0.42, 0.44	0.42, 0.44
400	1.34	0.56, 0.56	0.52, 0.53	0.50, 0.53	0.50, 0.58	0.50, 0.51	0.44, 0.45	0.41, 0.43	0.41, 0.43
450	1.34	0.55, 0.56	0.52, 0.53	0.50, 0.53	0.50, 0.57	0.49, 0.51	0.44, 0.45	0.41, 0.43	0.41, 0.43
500	1.34	0.55, 0.56	0.51, 0.52	0.50, 0.52	0.50, 0.56	0.50, 0.51	0.45, 0.46	0.42, 0.43	0.41, 0.43

Table 1: Comparative speed measurements for polyHash1271 and BRWHash1271 for different length (given in blocks) messages.

# blks	polyHash1305					BRWHash1305			
	$g = 1$	$g = 4$	$g = 8$	$g = 16$	$g = 32$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
1	2.31	2.38, 2.38	2.38, 2.38	2.25, 2.25	2.31, 2.31	2.06, 2.06	2.12, 2.12	2.19, 2.19	2.12, 2.12
2	1.69	1.84, 1.84	1.84, 1.84	1.84, 1.84	1.84, 1.84	1.22, 1.22	1.19, 1.19	1.22, 1.22	1.22, 1.22
3	1.56	1.48, 1.88	1.48, 1.88	1.48, 1.88	1.48, 1.88	0.96, 1.38	0.98, 1.35	0.98, 1.35	1.02, 1.38
4	1.52	1.38, 2.05	1.28, 1.94	1.28, 1.94	1.28, 1.92	1.16, 1.73	1.09, 1.62	1.08, 1.62	1.08, 1.62
5	1.49	1.40, 2.14	1.15, 1.94	1.15, 1.94	1.15, 1.94	0.93, 1.40	0.90, 1.35	0.90, 1.34	0.91, 1.34
6	1.47	1.23, 1.86	1.06, 1.95	1.06, 1.95	1.06, 1.98	0.81, 1.21	0.80, 1.17	0.81, 1.18	0.80, 1.18
7	1.46	1.16, 1.71	1.01, 1.98	1.01, 1.98	1.01, 1.99	0.81, 1.14	0.80, 1.11	0.80, 1.11	0.80, 1.12
8	1.45	1.15, 1.63	1.03, 2.09	0.97, 2.03	0.97, 2.02	0.91, 1.40	0.89, 1.38	0.89, 1.38	0.88, 1.38
9	1.47	1.19, 1.60	1.09, 2.11	0.94, 2.00	0.94, 2.01	0.81, 1.25	0.81, 1.23	0.81, 1.24	0.81, 1.24
10	1.43	1.12, 1.49	1.02, 1.94	0.91, 1.98	0.91, 1.99	0.78, 1.16	0.74, 1.14	0.76, 1.14	0.76, 1.14
11	1.45	1.07, 1.42	0.98, 1.83	0.88, 2.01	0.88, 2.01	0.76, 1.12	0.74, 1.10	0.76, 1.10	0.76, 1.10
12	1.47	1.07, 1.40	0.96, 1.73	0.87, 2.04	0.86, 2.03	0.83, 1.16	0.80, 1.13	0.80, 1.12	0.80, 1.12
13	1.48	1.11, 1.39	0.94, 1.65	0.85, 2.06	0.85, 2.06	0.77, 1.08	0.76, 1.05	0.76, 1.05	0.75, 1.05
14	1.47	1.06, 1.33	0.91, 1.57	0.84, 2.08	0.84, 2.08	0.74, 1.03	0.72, 1.00	0.72, 0.99	0.72, 0.99
15	1.47	1.03, 1.28	0.90, 1.51	0.82, 2.10	0.82, 2.10	0.74, 1.01	0.71, 0.97	0.72, 0.97	0.73, 0.97
16	1.46	1.03, 1.27	0.92, 1.50	0.85, 2.15	0.82, 2.12	0.79, 1.15	0.77, 1.12	0.76, 1.11	0.78, 1.12
17	1.40	1.06, 1.28	0.95, 1.49	0.89, 2.16	0.81, 2.08	0.75, 1.08	0.73, 1.06	0.72, 1.04	0.74, 1.06
18	1.40	1.03, 1.24	0.92, 1.43	0.86, 2.06	0.80, 2.10	0.72, 1.05	0.70, 1.02	0.70, 1.00	0.72, 1.01
19	1.41	1.00, 1.21	0.90, 1.39	0.85, 1.99	0.79, 2.11	0.74, 1.03	0.70, 1.00	0.69, 0.99	0.71, 1.00
20	1.41	1.01, 1.20	0.90, 1.36	0.84, 1.92	0.79, 2.12	0.78, 1.06	0.74, 1.02	0.73, 1.01	0.75, 1.02
21	1.41	1.03, 1.21	0.93, 1.32	0.85, 1.88	0.79, 2.12	0.74, 1.01	0.71, 0.98	0.70, 0.96	0.72, 0.98
22	1.41	1.05, 1.18	0.87, 1.29	0.83, 1.84	0.79, 2.15	0.72, 0.98	0.70, 0.95	0.68, 0.94	0.70, 0.95
23	1.41	0.99, 1.15	0.86, 1.26	0.88, 1.76	0.78, 2.17	0.73, 0.97	0.70, 0.94	0.68, 0.93	0.70, 0.93
24	1.40	0.99, 1.15	0.88, 1.26	0.87, 1.71	0.88, 2.19	0.76, 0.99	0.73, 0.97	0.71, 0.95	0.73, 0.96
25	1.40	1.01, 1.16	0.90, 1.27	0.89, 1.68	0.78, 2.17	0.73, 0.96	0.71, 0.93	0.69, 0.92	0.70, 0.93
26	1.40	0.99, 1.14	0.88, 1.24	0.88, 1.63	0.79, 2.20	0.72, 0.94	0.69, 0.91	0.68, 0.89	0.69, 0.90
27	1.41	0.98, 1.12	0.87, 1.22	0.85, 1.60	0.76, 2.19	0.72, 0.93	0.69, 0.90	0.68, 0.88	0.69, 0.90
28	1.40	0.98, 1.12	0.87, 1.20	0.80, 1.57	0.88, 2.23	0.75, 0.96	0.73, 0.92	0.71, 0.90	0.72, 0.92
29	1.40	1.00, 1.13	0.86, 1.18	0.79, 1.54	0.81, 2.24	0.72, 0.92	0.69, 0.89	0.68, 0.87	0.70, 0.89
30	1.40	0.98, 1.11	0.85, 1.16	0.78, 1.52	0.79, 2.24	0.72, 0.91	0.69, 0.87	0.67, 0.85	0.69, 0.87
31	1.40	0.97, 1.09	0.85, 1.14	0.85, 1.51	0.80, 2.25	0.72, 0.90	0.68, 0.87	0.67, 0.85	0.69, 0.87
32	1.40	0.97, 1.09	0.86, 1.14	0.82, 1.47	0.92, 2.31	0.75, 0.97	0.71, 0.95	0.70, 0.93	0.69, 0.92
33	1.40	0.99, 1.10	0.88, 1.15	0.81, 1.47	0.92, 2.32	0.72, 0.94	0.69, 0.92	0.68, 0.90	0.67, 0.89
34	1.40	0.97, 1.08	0.86, 1.13	0.86, 1.44	0.92, 2.27	0.71, 0.93	0.69, 0.90	0.67, 0.89	0.66, 0.88
35	1.39	0.96, 1.07	0.86, 1.12	0.80, 1.42	0.91, 2.22	0.71, 0.93	0.68, 0.90	0.68, 0.88	0.66, 0.88
36	1.39	0.97, 1.07	0.89, 1.11	0.79, 1.40	0.90, 2.18	0.74, 0.95	0.71, 0.91	0.69, 0.90	0.68, 0.89
37	1.39	0.98, 1.08	0.84, 1.09	0.79, 1.40	0.89, 2.14	0.72, 0.93	0.69, 0.89	0.68, 0.87	0.67, 0.87
38	1.39	0.97, 1.07	0.84, 1.08	0.79, 1.35	0.89, 2.10	0.71, 0.91	0.68, 0.88	0.67, 0.86	0.66, 0.85
39	1.39	0.96, 1.06	0.84, 1.07	0.79, 1.36	0.92, 2.07	0.71, 0.90	0.68, 0.87	0.67, 0.86	0.66, 0.85
40	1.39	0.96, 1.06	0.84, 1.07	0.81, 1.33	0.88, 2.04	0.73, 0.92	0.70, 0.89	0.69, 0.87	0.68, 0.86
41	1.39	0.97, 1.06	0.86, 1.08	0.84, 1.33	0.87, 2.00	0.71, 0.90	0.68, 0.87	0.67, 0.85	0.66, 0.84
42	1.39	0.96, 1.05	0.85, 1.07	0.83, 1.30	0.90, 1.97	0.71, 0.89	0.68, 0.86	0.67, 0.84	0.65, 0.83
43	1.39	0.95, 1.04	0.84, 1.06	0.78, 1.29	0.86, 1.94	0.71, 0.88	0.68, 0.85	0.67, 0.83	0.65, 0.83
44	1.39	0.96, 1.04	0.84, 1.05	0.78, 1.27	0.90, 1.92	0.73, 0.90	0.69, 0.87	0.68, 0.85	0.67, 0.84
45	1.39	0.97, 1.05	0.84, 1.04	0.80, 1.26	0.86, 1.89	0.72, 0.88	0.68, 0.85	0.67, 0.83	0.66, 0.82
46	1.39	0.96, 1.04	0.83, 1.03	0.77, 1.26	0.85, 1.86	0.71, 0.87	0.68, 0.84	0.66, 0.82	0.65, 0.82
47	1.38	0.95, 1.03	0.83, 1.02	0.83, 1.24	0.85, 1.84	0.71, 0.87	0.68, 0.83	0.66, 0.82	0.65, 0.81
48	1.39	0.95, 1.03	0.84, 1.03	0.79, 1.23	0.84, 1.82	0.73, 0.88	0.69, 0.85	0.68, 0.83	0.67, 0.82
49	1.39	0.96, 1.04	0.85, 1.04	0.79, 1.23	0.84, 1.79	0.71, 0.86	0.68, 0.83	0.67, 0.82	0.66, 0.81
50	1.38	0.95, 1.03	0.84, 1.02	0.79, 1.22	0.86, 1.79	0.70, 0.85	0.68, 0.82	0.66, 0.81	0.65, 0.80
100	1.37	0.94, 0.97	0.83, 0.91	0.77, 0.97	0.84, 1.29	0.71, 0.80	0.67, 0.77	0.66, 0.75	0.66, 0.75
150	1.37	0.93, 0.95	0.81, 0.87	0.76, 0.89	0.84, 1.14	0.71, 0.77	0.66, 0.74	0.65, 0.72	0.65, 0.72
200	1.36	0.93, 0.94	0.81, 0.85	0.75, 0.85	0.81, 1.03	0.71, 0.76	0.67, 0.72	0.65, 0.71	0.65, 0.71
250	1.36	0.93, 0.94	0.81, 0.85	0.76, 0.84	0.81, 0.98	0.70, 0.75	0.67, 0.71	0.65, 0.69	0.65, 0.69
300	1.36	0.93, 0.94	0.81, 0.84	0.76, 0.82	0.81, 0.95	0.71, 0.75	0.67, 0.71	0.65, 0.69	0.65, 0.69
350	1.36	0.92, 0.93	0.81, 0.83	0.75, 0.81	0.81, 0.94	0.70, 0.74	0.66, 0.70	0.65, 0.68	0.64, 0.68
400	1.36	0.92, 0.93	0.81, 0.83	0.74, 0.80	0.79, 0.91	0.70, 0.73	0.67, 0.70	0.65, 0.68	0.65, 0.68
450	1.36	0.92, 0.93	0.81, 0.82	0.75, 0.79	0.79, 0.89	0.70, 0.73	0.66, 0.69	0.65, 0.68	0.64, 0.67
500	1.36	0.92, 0.93	0.80, 0.82	0.74, 0.78	0.79, 0.88	0.70, 0.73	0.66, 0.69	0.65, 0.68	0.65, 0.67

Table 2: Comparative speed measurements for polyHash1305 and BRWHash1305 for different length (given in blocks) messages.

# blks	$t = 2$ $g = 4$	$t = 3$ $g = 4$	$t = 3$ $g = 8$	$t = 4$ $g = 4$	$t = 4$ $g = 8$	$t = 5$ $g = 4$	$t = 5$ $g = 8$
1	2.47, 2.47	2.47, 2.47	2.47, 2.47	2.47, 2.47	2.53, 2.53	2.53, 2.53	2.53, 2.53
2	1.37, 1.37	1.47, 1.47	1.40, 1.40	1.43, 1.43	1.43, 1.43	1.37, 1.37	1.40, 1.40
3	1.07, 1.87	1.04, 1.89	1.02, 1.87	1.09, 1.87	1.07, 1.87	1.07, 1.89	1.09, 1.89
4	1.37, 1.97	1.27, 1.80	0.90, 1.90	1.25, 1.80	0.90, 1.93	1.25, 1.80	0.90, 1.90
5	1.16, 1.61	1.08, 1.53	0.84, 1.83	1.09, 1.55	0.81, 1.81	1.08, 1.53	0.81, 1.83
6	1.03, 1.42	0.98, 1.36	0.76, 1.81	1.00, 1.37	0.76, 1.79	0.99, 1.36	0.76, 1.80
7	1.05, 1.39	1.02, 1.33	0.70, 1.56	1.04, 1.34	0.71, 1.58	1.03, 1.33	0.73, 1.57
8	0.97, 1.34	0.90, 1.27	0.89, 1.65	0.93, 1.35	0.96, 1.66	0.93, 1.29	0.85, 1.67
9	0.89, 1.22	0.83, 1.16	0.84, 1.49	0.87, 1.19	0.80, 1.51	0.86, 1.18	0.81, 1.53
10	0.84, 1.15	0.79, 1.08	0.79, 1.39	0.94, 1.12	0.77, 1.40	0.81, 1.11	0.77, 1.41
11	0.87, 1.15	0.91, 1.18	0.75, 1.30	0.86, 1.13	0.73, 1.30	0.85, 1.12	0.75, 1.32
12	0.81, 1.06	0.86, 1.10	0.72, 1.22	0.93, 1.06	0.71, 1.24	0.80, 1.05	0.72, 1.24
13	0.78, 1.01	0.82, 1.04	0.70, 1.16	0.77, 1.01	0.69, 1.17	0.77, 0.99	0.70, 1.18
14	0.74, 0.97	0.80, 1.00	0.69, 1.11	0.76, 0.97	0.72, 1.12	0.74, 0.95	0.68, 1.13
15	0.77, 0.98	0.81, 1.01	0.71, 1.10	0.85, 0.98	0.71, 1.13	0.78, 0.97	0.72, 1.14
16	0.75, 1.00	0.67, 0.92	0.67, 1.12	0.65, 0.89	0.65, 1.09	0.75, 0.99	0.68, 1.16
17	0.71, 0.96	0.64, 0.87	0.64, 1.06	0.62, 0.85	0.62, 1.04	0.72, 0.95	0.67, 1.11
18	0.70, 0.91	0.63, 0.84	0.63, 1.02	0.61, 0.82	0.61, 1.00	0.70, 0.92	0.69, 1.07
19	0.72, 0.92	0.71, 0.91	0.62, 0.98	0.69, 0.90	0.60, 0.97	0.73, 0.93	0.65, 1.04
20	0.67, 0.87	0.68, 0.88	0.61, 0.95	0.68, 0.86	0.59, 0.94	0.72, 0.91	0.63, 1.00
21	0.66, 0.87	0.67, 0.86	0.60, 0.94	0.66, 0.83	0.59, 0.92	0.70, 0.88	0.63, 0.98
22	0.65, 0.83	0.66, 0.84	0.59, 0.91	0.65, 0.82	0.58, 0.89	0.68, 0.86	0.62, 0.96
23	0.66, 0.84	0.68, 0.85	0.61, 0.92	0.67, 0.84	0.64, 0.95	0.71, 0.87	0.65, 0.97
24	0.64, 0.82	0.59, 0.76	0.58, 0.89	0.66, 0.82	0.63, 0.92	0.69, 0.85	0.62, 0.94
25	0.63, 0.81	0.57, 0.74	0.57, 0.86	0.65, 0.79	0.62, 0.90	0.68, 0.83	0.62, 0.92
26	0.62, 0.79	0.56, 0.72	0.57, 0.85	0.64, 0.78	0.61, 0.88	0.67, 0.82	0.61, 0.90
27	0.64, 0.80	0.62, 0.77	0.56, 0.82	0.66, 0.80	0.61, 0.86	0.69, 0.83	0.60, 0.88
28	0.62, 0.76	0.61, 0.76	0.56, 0.82	0.65, 0.78	0.60, 0.85	0.67, 0.81	0.60, 0.86
29	0.62, 0.75	0.61, 0.75	0.56, 0.80	0.64, 0.77	0.59, 0.83	0.66, 0.80	0.64, 0.85
30	0.60, 0.75	0.61, 0.73	0.55, 0.78	0.63, 0.76	0.59, 0.82	0.66, 0.78	0.59, 0.84
31	0.62, 0.75	0.63, 0.75	0.56, 0.80	0.65, 0.77	0.61, 0.84	0.68, 0.80	0.65, 0.85
32	0.61, 0.77	0.54, 0.70	0.54, 0.81	0.53, 0.69	0.53, 0.80	0.51, 0.67	0.51, 0.77
33	0.59, 0.75	0.53, 0.68	0.53, 0.79	0.52, 0.67	0.52, 0.78	0.50, 0.65	0.51, 0.75
34	0.59, 0.74	0.53, 0.68	0.53, 0.78	0.52, 0.67	0.52, 0.77	0.50, 0.65	0.50, 0.75
35	0.61, 0.75	0.57, 0.73	0.53, 0.77	0.57, 0.71	0.52, 0.76	0.55, 0.69	0.50, 0.73
36	0.58, 0.74	0.56, 0.71	0.52, 0.76	0.56, 0.70	0.51, 0.75	0.54, 0.69	0.50, 0.72
37	0.59, 0.72	0.56, 0.70	0.52, 0.76	0.55, 0.69	0.51, 0.74	0.54, 0.67	0.50, 0.72
38	0.58, 0.72	0.56, 0.69	0.52, 0.74	0.55, 0.68	0.51, 0.73	0.54, 0.67	0.50, 0.71
39	0.59, 0.72	0.58, 0.71	0.54, 0.75	0.57, 0.70	0.56, 0.77	0.56, 0.69	0.54, 0.75
40	0.58, 0.72	0.52, 0.65	0.52, 0.73	0.56, 0.69	0.54, 0.76	0.55, 0.67	0.53, 0.74
41	0.57, 0.70	0.51, 0.64	0.52, 0.72	0.55, 0.67	0.54, 0.74	0.54, 0.67	0.52, 0.73
42	0.57, 0.70	0.51, 0.63	0.52, 0.71	0.55, 0.67	0.53, 0.73	0.54, 0.66	0.52, 0.72
43	0.58, 0.70	0.55, 0.67	0.51, 0.71	0.57, 0.68	0.53, 0.73	0.56, 0.67	0.52, 0.71
44	0.57, 0.68	0.54, 0.66	0.51, 0.71	0.56, 0.67	0.56, 0.72	0.55, 0.67	0.52, 0.71
45	0.57, 0.68	0.53, 0.65	0.51, 0.69	0.55, 0.67	0.53, 0.71	0.55, 0.66	0.51, 0.70
46	0.56, 0.67	0.53, 0.65	0.51, 0.69	0.55, 0.66	0.52, 0.71	0.54, 0.66	0.51, 0.70
47	0.58, 0.70	0.58, 0.67	0.53, 0.69	0.57, 0.67	0.54, 0.72	0.56, 0.67	0.53, 0.71
48	0.56, 0.67	0.50, 0.60	0.51, 0.68	0.49, 0.60	0.49, 0.67	0.55, 0.66	0.52, 0.70
49	0.56, 0.66	0.50, 0.60	0.50, 0.68	0.49, 0.59	0.48, 0.66	0.55, 0.67	0.52, 0.69
50	0.56, 0.66	0.50, 0.60	0.51, 0.68	0.48, 0.59	0.48, 0.66	0.55, 0.65	0.52, 0.69
100	0.52, 0.58	0.48, 0.54	0.46, 0.56	0.47, 0.53	0.45, 0.55	0.46, 0.52	0.44, 0.54
150	0.51, 0.56	0.46, 0.51	0.45, 0.52	0.46, 0.50	0.43, 0.52	0.46, 0.51	0.46, 0.52
200	0.51, 0.54	0.45, 0.48	0.45, 0.51	0.45, 0.49	0.43, 0.50	0.44, 0.47	0.43, 0.49
250	0.51, 0.53	0.45, 0.47	0.45, 0.49	0.44, 0.47	0.43, 0.49	0.45, 0.48	0.43, 0.48
300	0.50, 0.53	0.45, 0.48	0.45, 0.51	0.45, 0.48	0.43, 0.48	0.44, 0.46	0.43, 0.47
350	0.50, 0.52	0.45, 0.47	0.44, 0.51	0.44, 0.46	0.43, 0.47	0.44, 0.46	0.43, 0.46
400	0.50, 0.52	0.44, 0.46	0.44, 0.50	0.42, 0.44	0.42, 0.46	0.43, 0.45	0.42, 0.45
450	0.50, 0.52	0.44, 0.46	0.44, 0.49	0.42, 0.44	0.42, 0.46	0.42, 0.44	0.42, 0.44
500	0.50, 0.51	0.44, 0.46	0.44, 0.49	0.43, 0.45	0.42, 0.46	0.43, 0.44	0.42, 0.45

Table 3: Speed measurements for t -BRWHash1271 for different length (given in blocks) messages.

# blks	$t = 2$ $g = 4$	$t = 3$ $g = 4$	$t = 3$ $g = 8$	$t = 4$ $g = 4$	$t = 4$ $g = 8$	$t = 5$ $g = 4$	$t = 5$ $g = 8$
1	2.50, 2.50	2.56, 2.56	2.50, 2.50	2.56, 2.56	2.56, 2.56	2.56, 2.56	2.50, 2.50
2	1.53, 1.53	1.53, 1.53	1.56, 1.56	1.56, 1.56	1.50, 1.50	1.53, 1.53	1.53, 1.53
3	1.21, 2.48	1.21, 2.50	1.23, 2.52	1.21, 2.54	1.21, 2.50	1.21, 2.50	1.21, 2.50
4	1.56, 2.47	1.48, 2.36	1.03, 2.61	1.45, 2.34	1.03, 2.62	1.45, 2.36	1.05, 2.64
5	1.35, 2.06	1.29, 2.00	0.95, 2.45	1.27, 2.00	0.94, 2.45	1.27, 1.99	0.95, 2.45
6	1.22, 1.84	1.19, 1.78	0.89, 2.39	1.18, 1.76	0.89, 2.38	1.17, 1.76	0.88, 2.38
7	1.21, 1.74	1.21, 1.71	0.85, 2.10	1.21, 1.71	0.85, 2.10	1.20, 1.71	0.84, 2.09
8	1.11, 1.75	1.11, 1.76	1.10, 2.42	1.11, 1.76	1.02, 2.33	1.11, 1.76	1.02, 2.33
9	1.06, 1.62	1.05, 1.62	1.05, 2.21	1.04, 1.62	0.98, 2.12	1.04, 1.61	0.96, 2.13
10	1.02, 1.52	0.99, 1.51	1.01, 2.05	1.00, 1.51	0.92, 1.99	1.00, 1.51	0.92, 1.97
11	1.02, 1.50	1.11, 1.58	0.95, 1.90	1.03, 1.50	0.90, 1.87	1.05, 1.49	0.89, 1.85
12	0.96, 1.41	1.05, 1.49	0.91, 1.79	0.99, 1.42	0.86, 1.74	0.99, 1.42	0.86, 1.74
13	0.94, 1.34	1.02, 1.40	0.89, 1.69	0.95, 1.35	0.83, 1.64	0.95, 1.35	0.83, 1.64
14	0.92, 1.28	0.99, 1.35	0.85, 1.60	0.93, 1.29	0.81, 1.56	0.93, 1.29	0.81, 1.57
15	0.94, 1.27	1.02, 1.35	0.90, 1.60	0.96, 1.30	0.85, 1.55	0.96, 1.30	0.94, 1.56
16	0.90, 1.33	0.88, 1.32	0.88, 1.65	0.87, 1.28	0.86, 1.62	0.93, 1.36	0.82, 1.59
17	0.88, 1.28	0.86, 1.26	0.85, 1.58	0.85, 1.23	0.85, 1.55	0.90, 1.31	0.81, 1.53
18	0.86, 1.24	0.84, 1.22	0.84, 1.53	0.83, 1.20	0.83, 1.51	0.89, 1.26	0.79, 1.47
19	0.88, 1.24	0.91, 1.28	0.83, 1.48	0.90, 1.25	0.81, 1.45	0.91, 1.27	0.78, 1.43
20	0.86, 1.20	0.89, 1.23	0.81, 1.43	0.88, 1.21	0.80, 1.40	0.89, 1.23	0.77, 1.38
21	0.85, 1.17	0.87, 1.21	0.80, 1.39	0.86, 1.18	0.79, 1.36	0.87, 1.20	0.76, 1.35
22	0.83, 1.14	0.86, 1.18	0.78, 1.34	0.85, 1.16	0.77, 1.32	0.86, 1.17	0.74, 1.31
23	0.85, 1.14	0.88, 1.19	0.81, 1.35	0.87, 1.17	0.84, 1.36	0.88, 1.18	0.83, 1.32
24	0.83, 1.11	0.80, 1.10	0.80, 1.33	0.86, 1.14	0.82, 1.33	0.87, 1.15	0.76, 1.27
25	0.82, 1.09	0.80, 1.07	0.79, 1.29	0.84, 1.11	0.81, 1.29	0.85, 1.12	0.79, 1.24
26	0.81, 1.07	0.79, 1.05	0.79, 1.27	0.84, 1.09	0.80, 1.27	0.84, 1.10	0.74, 1.21
27	0.83, 1.07	0.84, 1.10	0.77, 1.23	0.86, 1.10	0.79, 1.25	0.86, 1.11	0.74, 1.19
28	0.81, 1.05	0.83, 1.07	0.77, 1.21	0.85, 1.08	0.79, 1.22	0.85, 1.09	0.73, 1.17
29	0.81, 1.03	0.82, 1.06	0.76, 1.19	0.83, 1.06	0.77, 1.19	0.84, 1.07	0.72, 1.15
30	0.79, 1.02	0.81, 1.04	0.75, 1.16	0.82, 1.04	0.77, 1.17	0.83, 1.06	0.72, 1.13
31	0.81, 1.02	0.83, 1.06	0.77, 1.17	0.84, 1.05	0.78, 1.18	0.85, 1.07	0.74, 1.13
32	0.80, 1.06	0.78, 1.05	0.77, 1.22	0.76, 1.02	0.75, 1.19	0.74, 1.01	0.74, 1.18
33	0.79, 1.05	0.77, 1.03	0.76, 1.20	0.75, 1.01	0.74, 1.17	0.73, 0.99	0.74, 1.16
34	0.79, 1.03	0.77, 1.02	0.76, 1.18	0.74, 1.00	0.74, 1.15	0.73, 0.98	0.74, 1.14
35	0.79, 1.04	0.81, 1.06	0.75, 1.16	0.78, 1.03	0.73, 1.13	0.77, 1.02	0.73, 1.12
36	0.79, 1.02	0.80, 1.04	0.74, 1.14	0.77, 1.02	0.73, 1.12	0.76, 1.00	0.72, 1.11
37	0.78, 1.01	0.79, 1.03	0.74, 1.12	0.77, 1.00	0.72, 1.10	0.76, 0.99	0.71, 1.09
38	0.77, 1.00	0.78, 1.02	0.73, 1.11	0.76, 1.00	0.72, 1.09	0.75, 0.98	0.71, 1.07
39	0.79, 1.00	0.80, 1.02	0.75, 1.12	0.78, 1.00	0.75, 1.12	0.77, 0.99	0.75, 1.11
40	0.78, 0.99	0.75, 0.97	0.74, 1.11	0.77, 0.99	0.75, 1.10	0.76, 0.98	0.74, 1.09
41	0.77, 0.98	0.75, 0.96	0.74, 1.09	0.77, 0.97	0.74, 1.08	0.75, 0.96	0.73, 1.08
42	0.77, 0.97	0.74, 0.95	0.74, 1.08	0.76, 0.97	0.75, 1.08	0.75, 0.96	0.73, 1.07
43	0.78, 0.98	0.77, 0.98	0.73, 1.07	0.77, 0.98	0.73, 1.06	0.76, 0.97	0.73, 1.05
44	0.77, 0.96	0.77, 0.96	0.72, 1.05	0.77, 0.97	0.73, 1.05	0.76, 0.96	0.72, 1.04
45	0.77, 0.95	0.76, 0.96	0.72, 1.04	0.76, 0.96	0.72, 1.04	0.75, 0.95	0.72, 1.03
46	0.77, 0.94	0.76, 0.95	0.71, 1.03	0.76, 0.95	0.72, 1.03	0.75, 0.94	0.71, 1.02
47	0.77, 0.95	0.77, 0.96	0.73, 1.04	0.77, 0.96	0.73, 1.03	0.76, 0.95	0.73, 1.03
48	0.76, 0.94	0.73, 0.92	0.73, 1.04	0.71, 0.89	0.71, 1.01	0.76, 0.94	0.72, 1.01
49	0.76, 0.93	0.73, 0.91	0.72, 1.02	0.71, 0.88	0.71, 1.00	0.75, 0.93	0.72, 1.01
50	0.76, 0.92	0.73, 0.90	0.72, 1.01	0.71, 0.88	0.71, 0.99	0.75, 0.92	0.71, 1.00
100	0.73, 0.83	0.71, 0.81	0.69, 0.86	0.69, 0.79	0.67, 0.83	0.69, 0.79	0.67, 0.83
150	0.72, 0.80	0.69, 0.77	0.67, 0.81	0.68, 0.75	0.66, 0.78	0.68, 0.76	0.66, 0.78
200	0.72, 0.77	0.68, 0.74	0.67, 0.79	0.67, 0.73	0.66, 0.76	0.67, 0.73	0.66, 0.75
250	0.71, 0.76	0.68, 0.72	0.68, 0.76	0.66, 0.71	0.66, 0.74	0.67, 0.72	0.66, 0.73
300	0.71, 0.75	0.68, 0.72	0.67, 0.76	0.66, 0.71	0.66, 0.73	0.66, 0.71	0.65, 0.72
350	0.71, 0.74	0.67, 0.71	0.67, 0.74	0.66, 0.70	0.65, 0.72	0.66, 0.70	0.65, 0.71
400	0.71, 0.74	0.67, 0.70	0.67, 0.73	0.65, 0.69	0.65, 0.71	0.66, 0.69	0.65, 0.70
450	0.71, 0.73	0.67, 0.71	0.67, 0.73	0.65, 0.68	0.65, 0.70	0.65, 0.68	0.65, 0.69
500	0.71, 0.73	0.67, 0.70	0.67, 0.72	0.66, 0.68	0.65, 0.70	0.65, 0.68	0.65, 0.69

Table 4: Speed measurements for t -BRWHash1305 for different length (given in blocks) messages.

# blks	$d = 2$ $g = 4$	$d = 2$ $g = 8$	$d = 3$ $g = 4$	$d = 3$ $g = 8$	$d = 4$ $g = 4$	$d = 4$ $g = 8$	$d = 5$ $g = 4$	$d = 5$ $g = 8$
1	4.07, 5.00	4.07, 5.00	6.13, 6.13	6.13, 6.13	6.13, 6.13	6.20, 6.20	6.13, 6.13	6.13, 6.13
2	2.43, 3.37	2.43, 3.40	3.20, 3.60	3.20, 3.60	3.20, 3.60	3.20, 3.63	3.23, 3.60	3.20, 3.57
3	1.40, 1.93	1.31, 1.93	2.60, 2.93	2.27, 2.87	2.62, 2.93	2.27, 2.84	2.62, 2.93	2.27, 2.84
4	1.22, 1.70	1.17, 1.65	2.00, 2.23	1.77, 2.40	1.98, 2.25	1.77, 2.38	1.98, 2.23	1.77, 2.38
5	1.07, 1.61	1.05, 1.60	1.65, 2.04	1.49, 2.20	1.64, 2.04	1.51, 2.21	1.64, 2.04	1.48, 2.17
6	1.07, 1.44	1.04, 1.43	1.47, 1.90	1.30, 2.04	1.44, 1.88	1.32, 2.04	1.44, 1.88	1.30, 2.02
7	0.97, 1.33	0.96, 1.32	1.08, 1.46	1.08, 1.72	1.51, 1.87	1.37, 2.01	1.50, 1.88	1.36, 1.99
8	0.91, 1.23	0.89, 1.22	0.97, 1.38	0.99, 1.60	1.34, 1.65	1.23, 1.77	1.33, 1.64	1.22, 1.76
9	0.86, 1.19	0.83, 1.18	0.89, 1.26	0.95, 1.46	1.22, 1.50	1.13, 1.61	1.21, 1.49	1.13, 1.60
10	0.83, 1.15	0.80, 1.13	1.04, 1.27	0.91, 1.35	1.14, 1.41	1.05, 1.49	1.13, 1.39	1.05, 1.47
11	0.79, 1.08	0.76, 1.06	0.94, 1.19	0.84, 1.27	1.21, 1.44	1.00, 1.39	1.21, 1.43	0.99, 1.38
12	0.77, 1.10	0.73, 1.08	0.85, 1.08	0.82, 1.18	1.23, 1.33	0.96, 1.32	1.12, 1.32	0.95, 1.31
13	0.74, 1.07	0.71, 1.05	0.87, 1.06	0.78, 1.24	1.05, 1.25	0.92, 1.36	1.05, 1.24	0.91, 1.36
14	0.72, 1.02	0.70, 1.00	0.79, 1.03	0.79, 1.25	1.13, 1.20	0.89, 1.34	1.00, 1.19	0.88, 1.34
15	0.81, 1.07	0.68, 1.01	0.77, 0.98	0.76, 1.19	0.74, 0.96	0.77, 1.23	1.06, 1.23	0.94, 1.39
16	0.78, 1.03	0.66, 0.99	0.72, 0.95	0.73, 1.14	0.71, 0.93	0.76, 1.17	1.00, 1.16	0.89, 1.30
17	0.75, 0.99	0.65, 0.96	0.78, 0.99	0.72, 1.10	0.69, 0.90	0.72, 1.12	0.95, 1.11	0.86, 1.25
18	0.74, 0.95	0.64, 0.99	0.75, 0.95	0.69, 1.06	0.74, 0.95	0.70, 1.09	0.93, 1.07	0.83, 1.19
19	0.71, 0.93	0.63, 0.97	0.72, 0.91	0.69, 1.04	0.72, 0.91	0.69, 1.05	0.98, 1.11	0.82, 1.15
20	0.70, 0.91	0.62, 0.95	0.71, 0.90	0.67, 1.01	0.69, 0.88	0.70, 1.02	0.94, 1.06	0.80, 1.12
21	0.68, 0.86	0.61, 0.97	0.67, 0.87	0.70, 1.02	0.68, 0.87	0.67, 0.99	0.90, 1.02	0.78, 1.09
22	0.67, 0.85	0.61, 0.96	0.65, 0.84	0.65, 0.99	0.73, 0.91	0.72, 1.03	0.88, 1.00	0.77, 1.06
23	0.66, 0.83	0.60, 0.94	0.65, 0.82	0.65, 0.96	0.71, 0.87	0.70, 1.00	0.92, 1.03	0.81, 1.10
24	0.66, 0.85	0.59, 0.96	0.69, 0.88	0.64, 0.94	0.69, 0.85	0.68, 0.97	0.89, 0.99	0.79, 1.06
25	0.64, 0.84	0.58, 0.95	0.67, 0.84	0.63, 0.91	0.67, 0.83	0.67, 0.94	0.87, 0.97	0.77, 1.03
26	0.63, 0.82	0.58, 0.94	0.65, 0.81	0.62, 0.90	0.72, 0.86	0.66, 0.92	0.85, 0.95	0.75, 1.00
27	0.68, 0.85	0.62, 0.96	0.65, 0.80	0.62, 0.88	0.70, 0.84	0.65, 0.90	0.89, 0.98	0.74, 0.98
28	0.67, 0.84	0.62, 0.94	0.61, 0.80	0.62, 0.92	0.68, 0.82	0.65, 0.89	0.86, 0.95	0.73, 0.96
29	0.66, 0.83	0.61, 0.92	0.60, 0.78	0.61, 0.90	0.68, 0.81	0.64, 0.87	0.84, 0.93	0.72, 0.95
30	0.65, 0.81	0.60, 0.90	0.60, 0.77	0.60, 0.88	0.60, 0.73	0.64, 0.89	0.83, 0.91	0.71, 0.93
31	0.64, 0.80	0.59, 0.89	0.63, 0.80	0.59, 0.86	0.59, 0.72	0.63, 0.88	0.57, 0.71	0.57, 0.81
32	0.64, 0.79	0.59, 0.88	0.62, 0.78	0.59, 0.85	0.58, 0.71	0.63, 0.86	0.56, 0.70	0.56, 0.79
33	0.63, 0.76	0.58, 0.85	0.61, 0.77	0.59, 0.84	0.62, 0.74	0.62, 0.84	0.56, 0.69	0.56, 0.78
34	0.65, 0.76	0.57, 0.85	0.60, 0.75	0.58, 0.83	0.61, 0.73	0.62, 0.84	0.59, 0.72	0.55, 0.76
35	0.62, 0.75	0.57, 0.83	0.62, 0.76	0.58, 0.85	0.60, 0.71	0.61, 0.82	0.58, 0.71	0.55, 0.76
36	0.61, 0.74	0.57, 0.81	0.61, 0.75	0.57, 0.83	0.59, 0.70	0.61, 0.81	0.57, 0.69	0.55, 0.75
37	0.61, 0.73	0.56, 0.81	0.60, 0.74	0.57, 0.82	0.63, 0.74	0.64, 0.84	0.57, 0.69	0.54, 0.74
38	0.60, 0.72	0.56, 0.80	0.63, 0.77	0.56, 0.80	0.61, 0.72	0.62, 0.82	0.60, 0.71	0.58, 0.77
39	0.64, 0.75	0.56, 0.79	0.62, 0.76	0.56, 0.80	0.61, 0.71	0.62, 0.81	0.59, 0.70	0.56, 0.75
40	0.63, 0.74	0.56, 0.79	0.61, 0.74	0.56, 0.79	0.62, 0.70	0.61, 0.80	0.58, 0.69	0.56, 0.74
41	0.62, 0.74	0.55, 0.78	0.61, 0.73	0.56, 0.78	0.63, 0.73	0.61, 0.79	0.58, 0.68	0.56, 0.73
42	0.62, 0.73	0.58, 0.76	0.59, 0.71	0.56, 0.80	0.62, 0.71	0.61, 0.78	0.61, 0.71	0.55, 0.73
43	0.61, 0.72	0.55, 0.76	0.58, 0.70	0.55, 0.79	0.61, 0.71	0.60, 0.78	0.60, 0.70	0.55, 0.72
44	0.63, 0.71	0.55, 0.75	0.57, 0.69	0.55, 0.78	0.61, 0.70	0.60, 0.77	0.59, 0.69	0.55, 0.71
45	0.60, 0.70	0.54, 0.74	0.60, 0.71	0.55, 0.77	0.55, 0.66	0.59, 0.77	0.59, 0.68	0.55, 0.71
46	0.59, 0.70	0.54, 0.74	0.59, 0.70	0.54, 0.76	0.55, 0.65	0.59, 0.76	0.61, 0.71	0.57, 0.73
47	0.59, 0.69	0.54, 0.73	0.58, 0.69	0.54, 0.75	0.54, 0.65	0.58, 0.76	0.60, 0.69	0.56, 0.72
48	0.59, 0.71	0.54, 0.76	0.58, 0.69	0.54, 0.75	0.57, 0.67	0.58, 0.75	0.60, 0.69	0.58, 0.72
49	0.58, 0.71	0.56, 0.75	0.56, 0.67	0.54, 0.77	0.56, 0.66	0.58, 0.74	0.59, 0.68	0.56, 0.71
50	0.58, 0.70	0.56, 0.75	0.56, 0.66	0.54, 0.76	0.55, 0.65	0.58, 0.74	0.62, 0.70	0.55, 0.70
100	0.57, 0.62	0.53, 0.63	0.52, 0.58	0.50, 0.63	0.53, 0.58	0.54, 0.65	0.50, 0.57	0.50, 0.59
150	0.55, 0.58	0.51, 0.58	0.52, 0.56	0.49, 0.58	0.49, 0.54	0.53, 0.62	0.50, 0.55	0.49, 0.56
200	0.55, 0.56	0.51, 0.55	0.50, 0.53	0.48, 0.55	0.49, 0.52	0.51, 0.58	0.48, 0.52	0.47, 0.53
250	0.54, 0.55	0.50, 0.54	0.50, 0.52	0.49, 0.53	0.49, 0.51	0.51, 0.57	0.46, 0.50	0.46, 0.52
300	0.53, 0.55	0.50, 0.53	0.50, 0.52	0.48, 0.52	0.47, 0.50	0.51, 0.56	0.48, 0.51	0.47, 0.51
350	0.53, 0.54	0.50, 0.52	0.49, 0.51	0.48, 0.52	0.47, 0.49	0.51, 0.55	0.46, 0.49	0.46, 0.50
400	0.53, 0.54	0.50, 0.52	0.49, 0.51	0.48, 0.51	0.47, 0.49	0.51, 0.54	0.47, 0.49	0.46, 0.50
450	0.53, 0.54	0.49, 0.51	0.49, 0.50	0.47, 0.50	0.47, 0.49	0.51, 0.54	0.46, 0.48	0.45, 0.49
500	0.53, 0.54	0.49, 0.51	0.49, 0.50	0.47, 0.50	0.47, 0.49	0.51, 0.53	0.46, 0.47	0.45, 0.48

Table 5: Speed measurements for d -2LHash1271 for different length (given in blocks) messages.

# blks	$d = 2$ $g = 4$	$d = 2$ $g = 8$	$d = 3$ $g = 4$	$d = 3$ $g = 8$	$d = 4$ $g = 4$	$d = 4$ $g = 8$	$d = 5$ $g = 4$	$d = 5$ $g = 8$
1	4.25, 5.25	4.19, 5.19	5.75, 6.94	5.75, 6.94	5.75, 7.00	5.75, 7.00	5.75, 6.88	5.75, 7.00
2	2.38, 3.53	2.38, 3.72	3.09, 3.66	3.16, 3.72	3.12, 3.66	3.12, 3.69	3.12, 3.66	3.16, 3.66
3	1.44, 2.21	1.44, 2.46	2.58, 2.96	2.31, 3.06	2.56, 2.94	2.29, 3.08	2.56, 2.98	2.31, 3.06
4	1.28, 1.84	1.30, 2.02	2.09, 2.34	1.83, 2.91	2.08, 2.36	1.81, 2.89	2.09, 2.38	1.83, 2.92
5	1.16, 1.90	1.16, 2.09	1.76, 2.33	1.56, 2.70	1.86, 2.42	1.57, 2.70	1.76, 2.33	1.57, 2.69
6	1.17, 1.77	1.17, 1.91	1.56, 2.29	1.41, 2.70	1.56, 2.29	1.41, 2.71	1.56, 2.28	1.42, 2.71
7	1.11, 1.60	1.10, 1.75	1.23, 1.92	1.24, 2.46	1.54, 2.19	1.40, 2.56	1.54, 2.18	1.41, 2.55
8	1.04, 1.48	1.04, 1.60	1.12, 1.71	1.14, 2.20	1.42, 1.97	1.28, 2.28	1.41, 1.97	1.28, 2.27
9	1.01, 1.56	1.01, 1.63	1.12, 1.58	1.12, 2.02	1.38, 1.85	1.20, 2.09	1.32, 1.81	1.21, 2.08
10	0.98, 1.48	0.98, 1.54	1.11, 1.54	1.06, 1.88	1.24, 1.68	1.14, 1.93	1.24, 1.68	1.15, 1.94
11	0.95, 1.40	0.94, 1.45	1.09, 1.49	1.01, 1.75	1.32, 1.66	1.10, 1.81	1.26, 1.66	1.09, 1.81
12	0.97, 1.57	0.93, 1.57	1.04, 1.40	0.96, 1.64	1.21, 1.57	1.07, 1.70	1.20, 1.56	1.04, 1.70
13	0.95, 1.50	0.91, 1.51	1.00, 1.33	0.91, 1.72	1.22, 1.52	1.00, 1.81	1.15, 1.49	1.00, 1.80
14	0.92, 1.44	0.89, 1.44	0.96, 1.38	0.98, 1.83	1.11, 1.43	0.96, 1.79	1.11, 1.42	0.96, 1.79
15	0.97, 1.45	0.89, 1.50	0.94, 1.30	0.95, 1.75	0.92, 1.30	0.92, 1.65	1.12, 1.42	1.00, 1.78
16	0.95, 1.41	0.88, 1.45	0.93, 1.25	0.93, 1.67	0.89, 1.25	0.89, 1.59	1.09, 1.37	0.96, 1.69
17	0.93, 1.36	0.86, 1.40	0.95, 1.26	0.90, 1.60	0.88, 1.21	0.86, 1.52	1.06, 1.32	0.94, 1.62
18	0.92, 1.31	0.86, 1.46	0.92, 1.22	0.88, 1.54	0.91, 1.22	0.84, 1.47	1.03, 1.27	0.92, 1.57
19	0.90, 1.28	0.85, 1.42	0.90, 1.19	0.85, 1.48	0.88, 1.18	0.82, 1.42	1.05, 1.28	0.90, 1.51
20	0.89, 1.26	0.84, 1.38	0.89, 1.15	0.83, 1.43	0.89, 1.17	0.82, 1.37	1.03, 1.25	0.88, 1.46
21	0.88, 1.22	0.83, 1.46	0.86, 1.21	0.85, 1.50	0.87, 1.12	0.80, 1.33	1.01, 1.21	0.87, 1.42
22	0.88, 1.20	0.83, 1.43	0.85, 1.17	0.85, 1.46	0.89, 1.14	0.83, 1.34	0.99, 1.18	0.85, 1.37
23	0.86, 1.18	0.82, 1.39	0.83, 1.14	0.84, 1.41	0.87, 1.11	0.81, 1.30	1.00, 1.20	0.88, 1.38
24	0.88, 1.22	0.83, 1.47	0.85, 1.16	0.83, 1.38	0.88, 1.10	0.80, 1.27	0.99, 1.17	0.86, 1.34
25	0.87, 1.21	0.83, 1.44	0.84, 1.13	0.82, 1.34	0.85, 1.07	0.79, 1.24	0.97, 1.14	0.84, 1.31
26	0.86, 1.18	0.82, 1.41	0.83, 1.11	0.80, 1.31	0.90, 1.08	0.79, 1.21	0.95, 1.12	0.84, 1.28
27	0.89, 1.20	0.85, 1.42	0.82, 1.09	0.79, 1.28	0.86, 1.06	0.78, 1.19	0.97, 1.13	0.83, 1.25
28	0.88, 1.18	0.84, 1.39	0.83, 1.17	0.81, 1.35	0.87, 1.06	0.77, 1.16	0.96, 1.11	0.81, 1.23
29	0.87, 1.16	0.84, 1.36	0.82, 1.14	0.80, 1.32	0.84, 1.03	0.75, 1.14	0.94, 1.09	0.80, 1.20
30	0.86, 1.14	0.83, 1.34	0.80, 1.11	0.79, 1.29	0.78, 1.03	0.78, 1.20	0.93, 1.07	0.79, 1.18
31	0.86, 1.13	0.82, 1.32	0.82, 1.12	0.78, 1.27	0.77, 1.01	0.77, 1.18	0.76, 1.01	0.76, 1.17
32	0.85, 1.11	0.82, 1.30	0.82, 1.10	0.78, 1.25	0.77, 1.00	0.77, 1.16	0.75, 0.99	0.75, 1.15
33	0.85, 1.10	0.82, 1.28	0.80, 1.09	0.77, 1.22	0.78, 1.02	0.76, 1.14	0.75, 0.98	0.75, 1.13
34	0.84, 1.09	0.81, 1.26	0.80, 1.07	0.76, 1.20	0.78, 1.00	0.75, 1.12	0.76, 0.99	0.74, 1.11
35	0.84, 1.08	0.81, 1.25	0.82, 1.08	0.78, 1.26	0.78, 1.00	0.74, 1.11	0.76, 0.98	0.74, 1.10
36	0.85, 1.08	0.81, 1.23	0.81, 1.06	0.77, 1.24	0.76, 0.98	0.73, 1.09	0.76, 0.97	0.73, 1.08
37	0.87, 1.10	0.81, 1.22	0.80, 1.05	0.76, 1.22	0.78, 0.99	0.76, 1.10	0.75, 0.96	0.72, 1.06
38	0.84, 1.06	0.80, 1.20	0.81, 1.06	0.76, 1.20	0.78, 0.98	0.75, 1.08	0.77, 0.97	0.74, 1.08
39	0.86, 1.07	0.80, 1.19	0.81, 1.05	0.75, 1.19	0.79, 0.97	0.76, 1.07	0.76, 0.96	0.73, 1.06
40	0.85, 1.06	0.80, 1.18	0.80, 1.03	0.75, 1.17	0.76, 0.96	0.74, 1.06	0.76, 0.95	0.73, 1.05
41	0.85, 1.05	0.80, 1.17	0.80, 1.02	0.74, 1.15	0.78, 0.97	0.73, 1.04	0.75, 0.94	0.73, 1.04
42	0.84, 1.04	0.79, 1.16	0.78, 1.00	0.75, 1.20	0.78, 0.96	0.73, 1.03	0.77, 0.95	0.72, 1.02
43	0.84, 1.03	0.79, 1.15	0.78, 0.99	0.75, 1.18	0.78, 0.96	0.72, 1.02	0.76, 0.94	0.74, 1.01
44	0.83, 1.02	0.78, 1.13	0.77, 0.98	0.74, 1.16	0.77, 0.96	0.73, 1.00	0.76, 0.93	0.71, 1.00
45	0.83, 1.02	0.79, 1.13	0.78, 0.99	0.74, 1.15	0.73, 0.94	0.73, 1.05	0.75, 0.92	0.71, 0.99
46	0.83, 1.01	0.78, 1.12	0.78, 0.98	0.74, 1.14	0.73, 0.93	0.73, 1.04	0.77, 0.94	0.73, 1.00
47	0.82, 1.00	0.78, 1.11	0.78, 0.97	0.73, 1.13	0.72, 0.92	0.72, 1.03	0.76, 0.93	0.72, 0.99
48	0.83, 1.01	0.79, 1.13	0.77, 0.96	0.73, 1.11	0.74, 0.93	0.72, 1.02	0.76, 0.92	0.71, 0.98
49	0.83, 1.00	0.79, 1.12	0.76, 0.95	0.74, 1.17	0.73, 0.92	0.72, 1.01	0.76, 0.91	0.71, 0.97
50	0.82, 0.99	0.78, 1.11	0.76, 0.94	0.74, 1.16	0.74, 0.92	0.71, 1.00	0.77, 0.92	0.71, 0.96
100	0.81, 0.90	0.77, 0.94	0.73, 0.83	0.70, 0.94	0.70, 0.81	0.68, 0.87	0.69, 0.80	0.68, 0.84
150	0.81, 0.86	0.76, 0.87	0.72, 0.79	0.69, 0.86	0.68, 0.76	0.67, 0.83	0.69, 0.77	0.66, 0.79
200	0.80, 0.84	0.76, 0.84	0.71, 0.76	0.69, 0.81	0.68, 0.75	0.66, 0.78	0.67, 0.73	0.65, 0.76
250	0.80, 0.83	0.76, 0.82	0.71, 0.75	0.69, 0.79	0.67, 0.72	0.66, 0.76	0.66, 0.72	0.65, 0.76
300	0.80, 0.82	0.75, 0.81	0.71, 0.74	0.69, 0.77	0.67, 0.71	0.66, 0.74	0.67, 0.71	0.66, 0.74
350	0.80, 0.82	0.75, 0.80	0.71, 0.73	0.69, 0.76	0.67, 0.70	0.65, 0.72	0.66, 0.70	0.65, 0.72
400	0.80, 0.82	0.75, 0.79	0.71, 0.73	0.69, 0.75	0.67, 0.70	0.65, 0.72	0.66, 0.70	0.65, 0.71
450	0.79, 0.81	0.75, 0.79	0.70, 0.72	0.68, 0.74	0.66, 0.69	0.65, 0.71	0.66, 0.69	0.65, 0.71
500	0.79, 0.81	0.75, 0.78	0.70, 0.72	0.68, 0.73	0.67, 0.69	0.65, 0.70	0.65, 0.68	0.65, 0.70

Table 6: Speed measurements for d -2LHash1305 for different length (given in blocks) messages.

# bytes	BRWHash1271				BRWHash1305			
	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
10	3.20, 3.20	3.60, 3.60	3.50, 3.50	3.40, 3.40	3.30, 3.30	3.40, 3.40	3.50, 3.50	3.40, 3.40
50	1.20, 1.66	1.02, 1.46	1.02, 1.48	1.02, 1.46	1.48, 2.20	1.40, 2.10	1.40, 2.08	1.38, 2.08
100	0.67, 0.89	0.64, 0.85	0.63, 0.85	0.63, 0.84	0.92, 1.27	0.90, 1.24	0.90, 1.24	0.90, 1.25
500	0.55, 0.67	0.49, 0.61	0.47, 0.59	0.46, 0.58	0.76, 1.00	0.73, 0.97	0.72, 0.95	0.71, 0.94
1000	0.51, 0.58	0.45, 0.53	0.43, 0.51	0.43, 0.50	0.71, 0.83	0.68, 0.80	0.66, 0.78	0.66, 0.78
2000	0.51, 0.55	0.44, 0.49	0.43, 0.48	0.42, 0.47	0.70, 0.78	0.67, 0.74	0.65, 0.72	0.65, 0.72
3000	0.50, 0.53	0.44, 0.47	0.42, 0.45	0.42, 0.45	0.71, 0.77	0.67, 0.73	0.65, 0.71	0.65, 0.71
4000	0.50, 0.52	0.44, 0.46	0.42, 0.44	0.41, 0.44	0.70, 0.75	0.67, 0.71	0.65, 0.69	0.65, 0.69
5000	0.50, 0.52	0.44, 0.46	0.42, 0.45	0.41, 0.44	0.70, 0.74	0.67, 0.70	0.65, 0.69	0.65, 0.69
6000	0.50, 0.51	0.44, 0.45	0.41, 0.43	0.42, 0.43	0.70, 0.73	0.66, 0.70	0.65, 0.68	0.64, 0.68
7000	0.49, 0.51	0.44, 0.45	0.41, 0.43	0.41, 0.43	0.70, 0.73	0.66, 0.69	0.65, 0.68	0.65, 0.67
8000	0.49, 0.51	0.45, 0.47	0.42, 0.43	0.41, 0.43	0.70, 0.73	0.66, 0.69	0.65, 0.68	0.65, 0.67

Table 7: Comparison between BRWHash1271 and BRWHash1305 for different length (given in bytes) messages.

References

- [1] Sreyosi Bhattacharyya, Kaushik Nath, and Palash Sarkar. Polynomial hashing over prime order fields. *Advances in Mathematics of Communications*, 19(1):337–378, 2025.